Exercise Classification from Video

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Google Colab Link

Introduction

The goal of this project is to develop a deep learning model capable of classifying common home exercises from short video clips. The input to the model is a video clip approximately 3-10 seconds in length, represented as a sequence of frames, and the output is a predicted exercise label chosen from a set of 5 possible exercises. This project is useful because it can serve as a foundation for a larger system that can automatically track and count exercises from a live camera feed. The problem is challenging because the same exercise can look significantly different across people, environments, and camera angles, making machine learning well suited for the task as neural networks can learn complex visual and motion patterns directly from data.

Illustration

Figure 1 shows a high-level diagram of the project. The bubbles represent collections of data, while the rectangles represent black boxes that perform actions on a given input and produce an output.

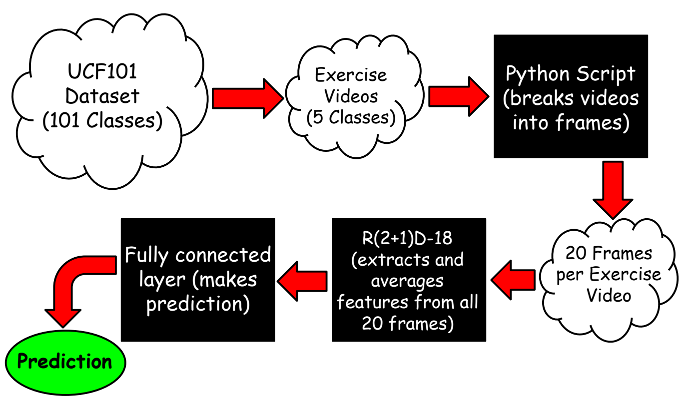

Figure 1: High-level diagram of the project.

Background & Related Work

Several previous projects explore video-based human action and exercise recognition. Riccio [2024] proposes a BiLSTM model that processes video frames alongside pose-based features to classify and count common strength exercises. Simonyan and Zisserman [2014] propose a two-stream CNN incorporating spatial and temporal networks, achieving strong action recognition performance on UCF101. Wang and Schmid [2013] track dense points through video frames and compute motion and appearance features along trajectories to recognize actions from UCF50. Hassan [2021] combines CNN features with LSTMs to learn temporal dependencies in action sequences, reporting competitive performance on UCF101. Liu et al. [2020] proposes a real-time 3D convolutional architecture that captures spatiotemporal features efficiently while reducing model size. Together these works show the progression from motion-pattern methods to deep learning approaches for video action recognition, motivating the use of CNN-based temporal modeling in this project.

Data Processing

Data comes from the UCF101 dataset [Soomro et al., 2012], a collection of 101 human action classes sourced from YouTube. A subset of 5 classes was selected, being bodyweight squats, jumping jacks, lunges, pushups, and wall pushups, totaling 594 videos with a fairly even class distribution as shown in Figure 2. Each video was processed by evenly sampling 20 frames, and an example frame is shown in Figure 3. Frames were stored as individual images in a separate folder for each video. UCF101 provides a predefined 75/25 train/test split ensuring no individual appears in both sets. Since no validation set is provided, the test set was manually split into validation and test subsets while preserving the same constraint, resulting in the distribution shown in Figure 4.

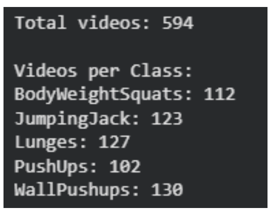

Figure 2: Total number of videos and class distribution that will be given to the model.

Figure 3: An example of one of the 20 frames extracted from a video in the dataset.

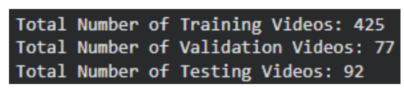

Figure 4: Distribution of training, validation, and testing data.

Architecture

The final model applies transfer learning using R(2+1)D-18 pretrained on the Kinetics-400 dataset as a backbone, with a custom classification head trained on the 5-class exercise subset of the UCF101 dataset. R(2+1)D-18 is an 18-layer video CNN that decomposes each 3D convolution into a 2D spatial convolution followed by a 1D temporal convolution [Tran et al., 2018]. The 2D convolution captures spatial appearance features within each frame, while the 1D temporal convolution captures motion patterns across frames. This decomposition allows the model to learn spatial and temporal features independently and is more parameter efficient than full 3D convolutions. All of the 20 frames in each video are preprocessed using the official torchvision transform provided with the pretrained R(2+1)D-18 weights, which resizes, crops, and normalizes the frames to match the format expected by the backbone [PyTorch Contributors, 2025]. The video is passed through the backbone, which outputs a 512-dimensional feature vector that then passes through a dropout layer with probability of 0.5, followed by a fully connected layer with ReLU activation, reducing the dimensionality to 256. Finally, it passes through another dropout layer and a final fully connected layer producing 5 class logits. The final model was trained using the Adam optimizer with a learning rate of 1e-4 and weight decay of 1e-3, using cross entropy loss for 10 epochs and batch size of 16.

Baseline Model

The baseline model is a simple 2D CNN that classifies a video based only on its first frame, treating the problem as standard image classification. It consists of three convolutional layers each followed by ReLU and max pooling, and two fully connected layers for classification. This baseline was chosen to gauge how much information a single frame contains for exercise classification. The model achieved 51.09% test accuracy, with similar exercises like pushups and wall pushups being frequently confused, suggesting that a single frame is insufficient for reliable exercise classification and highlighting the importance of temporal modeling across multiple frames.

Quantitative Results

The resultant accuracy on the test data of the final model was 98.91%. There is a large improvement from the previous models I constructed that did not utilize a pretrained model, to the final model that made use of the R(2+1)D-18 weights, as can be seen in Table 1. It is also clear from the plots in Figure 5, that the model is learning and is only slightly overfitting to the training data.

Model	**Brief Description**	**Test Accuracy**
Baseline	2D CNN that classifies based on 1st frame	51.09%
Primary	2D CNN that averages features over 32 frames	58.70%
Secondary	3D CNN that extracts spatiotemporal features across 20 frames	72.83%
Final	R(2+1)D-18 model as described above	98.91%

Table 1: Test accuracies of every model constructed for this project.

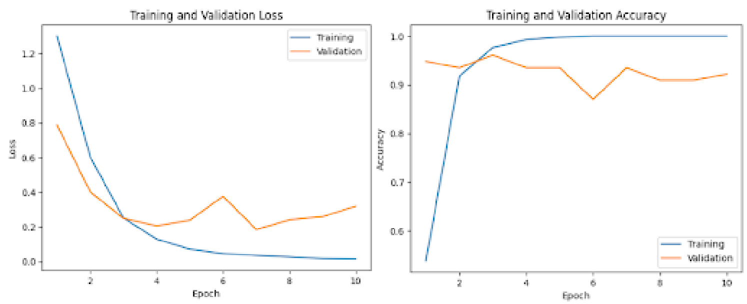

Figure 5: Training and validation loss and accuracy per epoch of the final model.

Qualitative Results

To better understand the model's performance, the per-class accuracies and confusion matrix are shown in Figures 6 and 7 respectively. The model performs perfectly on 4 out of 5 classes, with the only misclassification being a single bodyweight squats video being predicted as lunges. This error makes sense, as the up and down motion of bodyweight squats looks similar to the lunge motion. Overall, the consistent performance across classes suggests that the model has learned distinct visual representations for each exercise class.

```
Test Accuracy: 0.9891

Accuracy per Class:
BodyWeightSquats: 0.9412
JumpingJack: 1.0000
Lunges: 1.0000
PushUps: 1.0000
WallPushups: 1.0000
```

Figure 6: Overall and per-class accuracies of the final model.

Actual \ Predicted	BodyWeightSquats	JumpingJack	Lunges	PushUps	WallPushups
BodyWeightSquats	16	0	1	0	0
JumpingJack	0	21	0	0	0
Lunges	0	0	19	0	0
PushUps	0	0	0	17	0
WallPushups	0	0	0	0	18

Figure 7: Confusion matrix of the final model's predictions.

Evaluation on New Data

The test set that achieved the results above was not seen by the model at all during training and hyperparameter tuning, and all model decisions were based solely on the model's performance with respect to the validation set. As mentioned above, the test and validation split ensures no same person appears in both sets, suggesting the results are a reliable measure of generalization to unseen data. However, since R(2+1)D-18 was pretrained on Kinetics-400, which similarly sources videos from YouTube [Kay et al., 2017], there is a small possibility of overlap between the pretraining data and the UCF101 test set. For this reason I recorded my own custom test videos of all 5 exercises, which I can guarantee the model has never seen. For each class, I recorded across 2 different backgrounds, lighting conditions, and outfits, and performed each exercise facing 4 different directions, resulting in the balanced distribution shown in Figure 8. The videos were resized, then processed identically to the UCF101 data. The overall accuracy on the custom test set was 65%, with per-class accuracies shown in Figure 9. The model completely failed on lunges, always predicting wall pushups, which makes sense as most UCF101 lunge videos show the exercise performed with dumbbells, which the model likely learned as a distinct feature, whereas my videos did not include dumbbells. The model performed well on the remaining classes, with the exception of bodyweight squats, which may be attributed to a similar dataset-specific bias absent in my recordings.

```
Number of Custom Test Videos: 40

Videos per Class:
BodyWeightSquats: 8
JumpingJack: 8
Lunges: 8
PushUps: 8
WallPushups: 8
```

Figure 8: Total number of videos and class distribution of the custom test data.

```
Test Accuracy: 0.6500

Accuracy per Class:
BodyWeightSquats: 0.5000
JumpingJack: 0.8750
Lunges: 0.0000
PushUps: 0.8750
WallPushups: 1.0000
```

Figure 9: Overall and per-class accuracies of the final model on the custom test data.

Discussion

Overall, the final model performs excellently on UCF101 but surprisingly struggles on the custom videos, suggesting that high test accuracy may not translate to real world performance. The non-uniform per-class results indicate that the issue stems from specific dataset biases rather than a general inability to classify exercises, as seen by the lunge failure above. This demonstrates that models can learn specific visual features present in training data rather than general patterns that define the action, which leads to inaccuracies when those features are absent. The key takeaway is that accuracy on a single dataset is insufficient for evaluating a model intended for real world use, and that training data must be representative of the environment in which the model will be used.

Ethical Considerations

As mentioned earlier, this model could be used in real-time exercise tracking, where a camera would monitor a person's workout, counting how many times they perform a specific exercise. This could raise ethical concerns about privacy and consent, since cameras could monitor people's workouts and their information could be used to train the model. Additionally, the model has limitations as it may not generalize well to different environments, camera angles, or body types, and its performance could decrease when processing data that is significantly different from the training data.

Project Difficulty / Quality

Video classification is inherently more challenging than image classification, as the model must learn to extract features across both spatial and temporal dimensions rather than a single static frame. This is shown by the baseline's 51.09% accuracy when classifying based on a single frame. An additional difficulty specific to exercise classification is the variance in visual features within the same class. For example, the same exercise may look significantly different if a person has different form, speed, or body type. Despite these difficulties, the final model achieves 98.91% on the UCF101 test set, demonstrating strong performance on this difficult problem.

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